

Vinayak Koli

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EDUCATION

Indraprastha Institute of Information Technology Delhi (IIIT-D)

2023 – 2027

B.Tech., Computer Science and Social Sciences

Delhi, India

- **Coursework:** Data Structures & Algorithms, Algorithm Design, DBMS, OOP, Operating Systems, Collaborative Filtering

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, SQL, PHP

AI / ML: PyTorch, LightGBM, LangChain, HuggingFace Transformers, Ollama, LanceDB, NetworkX, Gradio, Optuna

Frameworks & Tools: React Native, FastAPI, Electron, Laravel, Vite, Zustand, Git, Appwrite, Expo, MySQL, Postman

EXPERIENCE

Undergraduate Researcher – AI Model Evaluation

Jan 2026 – Present

IIIT Delhi | Prof. Suman Roy

Delhi, India

- Designed a **7-step ontology-guided LLM code generation pipeline**: ontology retrieval (9-category knowledge graph), structured plan generation, self-scoring gate, hint-injected plan revision, sandboxed execution, iterative repair loop (3 iterations with stagnation detection), and adaptive graph edge-weight updates on task success.
- Benchmarked **10 local LLMs** (CodeLlama, Qwen2.5, Qwen2.5-Coder 7B–32B via Ollama) on OpenAI HumanEval (164 problems) across Direct, SCoT, and Ontology-Guided strategies; achieved **76.22% pass@1** (+11.6pp over baseline); identified self-evaluation bias (plan-success $r=0.168$) and Qwen2.5-Coder collapse at 14B/32B scale.
- Previously built a custom evaluation system for the **DeepMind Code Contests** dataset (165+ problems) with automated execution, correctness checking, and structured pass@k / failure-mode logging across multiple SLMs.

Web Development Back-End Intern

Nov 2024 – Jan 2025

1Stop (World Risk Governance)

Remote

- Built a **RESTful Task Manager API** (Laravel 11, full CRUD, Blade frontend) and a full-stack **Booking Management System** with role-based dashboards (Admin/User), Laravel Breeze authentication, and CMS-driven pages.
- Designed and migrated **MySQL schemas** for both systems; applied Eloquent ORM, middleware-protected route groups, and server-side deduplication logic.

PROJECTS

Hybrid Movie Recommendation System | Python, PyTorch, LightGBM, LangChain, Gradio, Optuna | [GitHub](#) | [Live Demo](#)

- Built a movie recommendation system combining **7 AI models** (collaborative filtering, content-based, matrix factorisation, LLM-augmented) trained on **1M+ user ratings**, ensuring no single model's blind spots affect the final output.
- Leveraged a large language model to extract semantic user preference signals (likes/dislikes reasoning) and blended them with traditional recommendation signals via a **LightGBM meta-learner** for personalised re-ranking.
- Achieved **80% hit rate** with near-perfect personalisation scores and strong catalogue coverage; tuned hyperparameters with **Optuna** and deployed an interactive demo via **Gradio**.

Context Lens Desktop | Electron, React, TypeScript, Ollama, LanceDB, OpenAI API | [GitHub](#) | [Live Demo](#)

- Built a cross-platform desktop AI assistant using **Electron 32 + React 18** with global OS-level text-selection detection via native hooks (`uihook-napi`), triggering a floating chat panel with zero copy-paste friction.
- Designed a unified token-streaming IPC architecture supporting **NDJSON** (Ollama), **SSE** (OpenAI), and **Anthropic** event streams simultaneously, with hardware-aware local model auto-selection based on detected RAM.
- Built a fully **on-device RAG pipeline** (LanceDB + `nommic-embed-text`): PDF/Markdown/text ingestion with word-boundary chunking, vector search, and per-reply source citations – all data remaining on-device.
- Implemented persistent **conversational memory** via a separate LanceDB vector table; parallelised memory recall and RAG search using `Promise.all`, eliminating sequential embedding round-trips to the local Ollama server.

Vocal for Local (VoLo) | React Native, FastAPI, Python, Appwrite | [LinkedIn](#) | [Live Demo](#)

- Built a cross-platform mobile app (iOS & Android) for local business discovery with Google Sign-In, location-based search across name/category/location, and a 14-screen UX built from scratch; implemented custom walking-distance calculation without any third-party mapping API.
- Improved app response speed by **~35%** via server-side caching of frequent requests and progressive UI rendering; backend powered by **FastAPI** with **Appwrite** for authentication and storage.

ACHIEVEMENTS & ACTIVITIES

- **3rd Place** – DevHaven 2.0 Hackathon (Competitive Programming & AI Track)
- **Captain**, Table Tennis Team, Karwaan 2024 | Member, Sports Council Operations Team, IIIT Delhi
- Led a **10+ member team** delivering a cancer awareness workshop impacting **100+ individuals** at Tihar Jail